

Homework 8:

Low-Fidelity Prototypes

Sofia Hirao, Jolie Abadir, Sam

All Topics

Sofia: Basics of piping techniques how different tips create different designs

Jolie: Reading a squat assessment — identifying good vs. compensated form

Sam:

Domain: Baking

Topic: Basics of piping techniques how different tips create different designs

This topic meets the requirements because it is a focused, visual skill that can be taught in 10 minutes through demonstration and practice.

Expert: Sofia

- Jolie: Has baked before and used a piping tip but has struggled in the past.
- Sam: does not really bake and is not familiar with these terms and usages.

Media: What media would you use?

- 1) Step-by-step video demonstrations of each piping tip in action.
- 2) Pictures showing the final designs for each tip.
- 3) short animated GIFs showing the pressure and motion for piping.

What is one positive piece of feedback your TA gave you on this idea?

- This is something teachable in 10 minutes and it is something someone would want to know

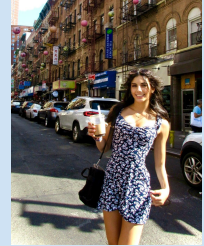
What is one negative or cautionary piece of feedback your TA gave you on this idea?

- This is something that requires following along and feedback to make sure they are doing it correctly so keeping that in mind when thinking about this topic.

User:

- Jolie Abadir is a current sophomore at Barnard College, majoring in Computer Science.
- Jolie is interested in learning how to use different piping tips because she wants to bake and decorate cakes for her friends' birthdays. For example, she recently wanted to make cupcakes for a friend's party but couldn't create decorative frosting patterns, so learning proper piping techniques would let her create visually appealing desserts.

PICTURE



Quiz:

- Match the piping tip to its frosting design
- Identify how much pressure is required for each pattern
- Simulate piping a shape using a drag and drop or hover over activity to practice technique

Who will you run your prototype on?

- I will run the prototype with another student in class who has minimal baking experience and is interested in learning cake decorating.

Domain: Movement Analysis & Biomechanics

Topic: Reading a squat assessment — identifying good vs. compensated form

This topic meets the requirements because squat assessments have clear visual markers (knee cave, forward lean, heel rise) that can be taught and quizzed in 10 minutes using side-by-side video comparison.

Expert: Jolie

- Sofia: Has intuitive body awareness from soccer and running but lacks formal movement analysis vocabulary.
- Sam: Recreational experience in athletics but no formal movement analysis background

Media: What media would you use?

- 1) Side-by-side videos of good vs. compensated squats, paused at key checkpoints.
- 2) Annotated images highlighting compensation patterns (knee cave, forward lean, heel rise).
- 3) Slow-motion video clips with visual overlays marking joint alignment.

Positive TA Feedback: This seems like a viable topic that could work well as a lesson.

Negative TA Feedback: For anything physical, it may be hard to teach virtually — need to find ways to make it work without hands-on correction.

User:

- Sofia Hirao is a 19-year-old sophomore at Barnard College. She is a runner and former soccer player with intuitive body awareness but no formal movement analysis training.
- Sofia is interested in learning movement analysis to prevent injuries and improve her athletic performance. She has noticed her knees caving inward while running but couldn't identify it as a compensation pattern without guidance.



Quiz:

- Watch a squat video and identify whether it shows good or compensated form.
- Label which compensation pattern is present (knee cave, forward lean, heel rise).
- Given two squat videos, identify which person has better form and explain why.

Who will you run your prototype on?

I would run this one another student in class without formal movement analysis experience.

Domain: Skincare

Topic: Reading a squat assessment — identifying good vs. compensated form

This topic meets the requirements because squat assessments have clear visual markers (knee cave, forward lean, heel rise) that can be taught and quizzed in 10 minutes using side-by-side video comparison.

Expert: Jolie

- Sofia: Has intuitive body awareness from soccer and running but lacks formal movement analysis vocabulary.
- Sam: Recreational experience in athletics but no formal movement analysis background

Media: What media would you use?

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